



Activity 4

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Part A: Q1: Predict output



```
int arr[4] = {10, 20, 30, 40};  
int i;  
  
for(i = 0; i < 4; i++)  
{  
    printf("%d ", arr[i]);  
}
```

Output

10 20 30 40

=== Code Execution Successful ===

Q2: Predict output



```
int arr[3] = {5,10,15};  
int i;  
  
for(i = 2; i >= 0; i--)  
{  
    printf("%d ", arr[i]);  
}
```

Output

15 10 5

Q3. Predict output



```
int arr[2][2] = {{1,2},{3,4}};  
printf("%d", arr[1][0]);
```

Output

3

Q4. Predict output

```
int arr[2][2] = {{1,2},{3,4}};  
int i, j, sum = 0;  
  
for(i = 0; i < 2; i++)  
{  
    for(j = 0; j < 2; j++)  
    {  
        sum += arr[i][j];  
    }  
}  
  
printf("%d", sum);
```

Output
10



Q5. Predict output



```
int arr[3] = {10,20,30};  
printf("%d", arr[1+1]);
```

Output

30

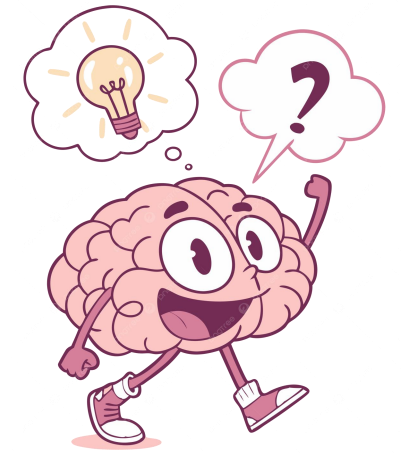
Part B: Q1. Correct the Code



```
int arr[3] = {1,2,3};  
printf("%d", arr[3]);
```

✗ Problem: index out of range
arr[2]

Q2. Correct the Code



```
int arr[3];  
int i;  
  
for(i = 0; i < 3; i++)  
{  
    scanf("%d", arr[i]);  
}
```

```
scanf("%d", &arr[i]);
```


Q3. Correct the Code

```
int arr[2][2] = {{1,2},{3,4}};  
int i, j;  
  
for(i = 0; i < 2; i++)  
{  
    for(j = 0; j < 3; j++)  
    {  
        printf("%d ", arr[i][j]);  
    }  
}
```

✗ Problem: array has only 2 columns
for (j = 0; j < 2, j++)



Q4. Correct the Code



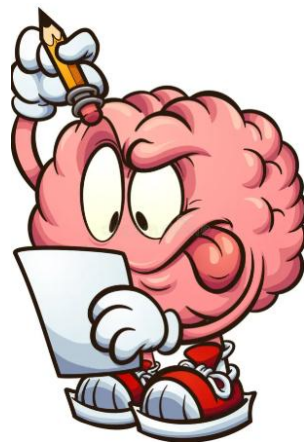
```
int arr[2][2] = {1,2,3,4};  
printf("%d", arr[0][1]);
```

X Wrong Initialization

```
int arr[2][2] = {{1,2},{3,4}};
```

BONUS Question

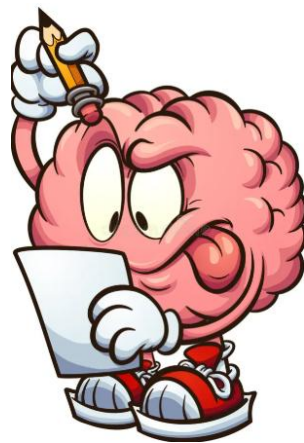
- Write a C program that:
 - Takes input for a 2×3 matrix
 - Prints:
 - All elements
 - Sum of all elements
 - Largest element



```
1  #include <stdio.h>
2  int main()
3  {
4      int arr[2][3];
5      int sum = 0, max;
6      printf("Enter elements:\n");
7      for(int i = 0; i < 2; i++)
8      {
9          for(int j = 0; j < 3; j++)
10         {
11             scanf("%d", &arr[i][j]);
12         }
13     }
```

BONUS Question

- Write a C program that:
 - Takes input for a 2×3 matrix
 - Prints:
 - All elements
 - Sum of all elements
 - Largest element



```
14     max = arr[0][0];
15     printf("Matrix:\n");
16     for(int i = 0; i < 2; i++)
17     {
18         for(int j = 0; j < 3; j++)
19         {
20             printf("%d ", arr[i][j]);
21             sum += arr[i][j];
22             if(arr[i][j] > max)
23                 max = arr[i][j];
24         }
25         printf("\n");
26     }
27     printf("Sum = %d\n", sum);
28     printf("Max = %d\n", max);
29     return 0;
30 }
```